CHAPTER THREE

METHODOLOGY

3.1 Introduction

This chapter presents the the theory of the multiple linear regression model as well as the Learst square principal solution of the model as demonstrated in the following subsections.

3.2 Multiple linear regression

Multiple linear regression allows us to investigate the joint effect of two or more predictors on the response; i.e. relate a single response variable to two or more predictors simultaneously. The predictor variables may be all quantitative or both quantitative and categorical.
With, multile linear regression model we can ;

(i) show how strong the relationship is between two or more predictor variables and one response variable.

(ii) predict the value of the response variable at a certain value of the predictor variables.

3.3 Multiple linear regression model

Suppose we want to use k predictor variables $X_1, X_2, \ldots, X_k$ to explain a response variable y then the model is of the form:

$$y = \beta_0 + \beta_1 X_1 + \ldots + \beta_k X_k + \epsilon \tag{3.1}$$

Given a random sample of size n selected from the population, the model is of the form

$$Y_i = \beta_0 + \beta_1 X_{i1} + \ldots + \beta_k X_{ik} + \epsilon_i; \quad i = 1, 2, \ldots, n \tag{3.2}$$

In matrix form we have

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & & & & \vdots \\ 1 & x_{n1} & X_{n2} & \cdots & x_{nk} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

i.e

$$\underbrace{\mathbf{y}}_{n\times 1} = \underbrace{\mathbf{X}}_{n\times(k+1)} \underbrace{\underline{\beta}}_{(k+1)\times 1} + \underbrace{\epsilon}_{n\times 1} \tag{3.3}$$

where;
The matrix $\mathbf{X}$ is called design matrix ;
$\underline{\beta}$ is a column vector of regression coefficients.
$\mathbf{y}$ is a column of response values for individuals in the sample.
ϵ_i is the error term of individual observations of the outcome y_i

Assumptions of the model:

- Each error term in the error vector has mean zero; $E[\underline{\epsilon}] = \underline{0}$
- The error terms are independent of each other; Cov $(\epsilon_i, \epsilon_j) = 0$
- The error terms have constant variance; $E\,[\underline{\epsilon\epsilon}' = \sigma^2 I_n$
- The error terms are normally distributed with mean 0 and finite variance σ^2.
- The response variable is normally distributed.
- There is no correlation between the error terms and the predictor variables.

3.4 Estimation of the Model Parameters

In order to estimate the model parameters $\beta_0, \beta_1, \ldots, \beta_p$ in 3.1 we use the principle of least squares which minimizes the error sum os squares(SSE). Suppose that $n > k$ observations are available, and let y_i denote the $i^t h$ observed response and x_{ij} denote the $i_t h$ observation. We assume that the error term ε in the model has $E(\varepsilon) = 0, Var(\varepsilon) = \sigma^2$, and that the errors are uncorrelated.
The general model of the Learst squares function is

$$S\left(\beta_0, \beta_1, \ldots, \beta_k\right) = \sum_{i=1}^{n} \varepsilon_i^2 = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij}\right)^2 \tag{3.4}$$

The function S must be minimized with respect to $\beta_0, \beta_1, \ldots, \beta_k$. The least-squares estimators of $\beta_0, \beta_1, \ldots, \beta_k$ must satisfy

$$\left.\frac{\partial S}{\partial \beta_0}\right|_{\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_k} = -2 \sum_{i=1}^{n} \left(y_i - \hat{\beta}_0 - \sum_{j=1}^{k} \hat{\beta}_j x_{ij}\right) = 0 \tag{3.5}$$

and

$$\left.\frac{\partial S}{\partial \beta_j}\right|_{\hat{k}_0\hat{\beta}_1\ldots\hat{\beta}_k} = -2\sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \sum_{j=1}^{k}\hat{\beta}_j x_{ij}\right)x_{ij} = 0, \quad j = 1, 2, \ldots, k \tag{3.6}$$

Simplifying equation 3.6, we obtain the least-squares normal equations

$$n\hat{\beta}_0 + \hat{\beta}_1\sum_{i=1}^{n}x_{i1} + \hat{\beta}_2\sum_{i=1}^{n}x_{i2} + \cdots + \hat{\beta}_k\sum_{i=1}^{n}x_{ik} = \sum_{i=1}^{n}y_i \tag{3.7}$$

which simplifies to

$$\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{Y} \tag{3.8}$$

$$SSE = (\mathbf{Y} - \mathbf{X}\underline{\beta})'(\underline{\mathbf{Y}} - \mathbf{X}\underline{\beta})$$

Differentiating SSE wrt to each unknown regression coefficient yields a $(k+1)$ simultaneous equations which are called normal equations.
The matrix of the normal equation is given

$$X'\underline{Y} = (X'X)\underline{\beta} \tag{3.9}$$

The solution is given by;

$$\underline{\beta} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_j \\ \vdots \\ y_n \end{bmatrix} = (X'X)^{-1}X'Y$$

provided the inverse matrix $(X'X)^{-1}$ exists.

Fitted values

Let $\hat{\beta}$ be the estimator of β for the model $y = X\beta + \epsilon$, we then define $\hat{y} = X\beta$ where β is any estimator of β R^2 is the percentage of the dependent variable explained by the independent variable. The coefficient of multiple determination R^2 is

$$R^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$$

Multiple regression explains most of the observed y variation using a model with relatively few predictors that are easily interpreted. We therefore adjust R^2 to the size of the model

$$R_a^2 = 1 - \frac{n-1}{n-(p+1)} \times \frac{SSE}{SST}$$

and since the ratio infront of $\frac{SSE}{SST}$ exceeds $1, R_a^2$ is smaller than R^2. The larger the number of predictors p relative to the sample size n, the smaller R_α^2 will be relative to R^2. When R^2 is adjusted, it can take a negative value while R^2 itself must be between 0 and 1 . If R_a^2 is smaller than R^2, then the model may contain many predictors.

Properties of Learst Squares Estimators

- $\hat{\beta}$ is unbiased estimator of β if the model is correct.
- $cov(\beta) = var(\hat{\beta} = var[(X'X)^{-1}X'Y]$

Estimation of σ^2

The method of least squares cannot be used to estimate σ^2 because σ^2 does not appear in $s(\beta)$ We use the SSE as the basis for estimating

$$\hat{\sigma}^2 = s^2 = \frac{SSE}{n-(p+1)}$$

3.5 HYPOTHESIS TESTING IN MULTIPLE LINEAR REGRESSION

Once we have estimated the parameters in the model, we face two immediate questions:

1. What is the overall adequacy of the model?
2. Which specific regressors seem important?

Several hypothesis testing procedures prove useful for addressing these questions. The formal tests require that our random errors be independent and follow a normal distribution with mean $E(\epsilon_i) = 0$ and variance $Var(\epsilon_i) = \sigma^2$

3.5.1 Test for Significance of Regression

The test for significance of regression is a test to determine if there is a linear relationship between the response y and any of the regressor variables $x_1, x_2, \ldots, x_k$
This procedure is often thought of as an overall or global test of model adequacy.
The appropriate hypotheses are

$$H_0 : \beta_1 = \beta_2 = \cdots \beta_k = 0$$

$$H_1 : \beta_j \neq 0 foratleastj$$

Rejection of this null hypothesis implies that at least one of the regressors $x_1, x_2 \ldots x_k$ contributes significantly to the model.
The test procedure is a generalization of the analysis of variance used in simple linear regression. The total sum of squares SST is partitioned into a sum of squares due to regression, SSR, and a residual sum of squares, SSE. Thus,
This shows that if the null hypothesis is true,then $SSR_R \div \sigma$ follows a χ^2_k distribution, which has the same number of degrees of freedom as number of regressor variables in the model.

$$F_0 = \frac{SS_{\mathrm{R}}/k}{SS_{\mathrm{Res}}/(n-k-1)} = \frac{MS_{\mathrm{R}}}{MS_{\mathrm{Res}}} \tag{3.10}$$

follows the $F_{k,n-k-1}$ distribution.
Suppose the expected mean squares indicate that if the observed value of F_0 is large, then it is likely that at least one $\beta_j \neq 0$ therefore we reject the null hypothesis.
The test procedure is usually summarized in an analysis-or-variance

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
Regression	SS_{R}	k	MS_{R}	$MS_{\mathrm{R}}/MS_{\mathrm{Res}}$
Residual	SS_{Res}	$n-k-1$	MS_{Res}	
Total	SS_{T}	$n-1$		

and since

$$SS_{\mathrm{T}} = \sum_{i=1}^{n} y_i^2 - \frac{\left(\sum_{i=1}^{n} y_i\right)^2}{n} = \mathbf{y}'\mathbf{y} - \frac{\left(\sum_{i=1}^{n} y_i\right)^2}{n} \tag{3.11}$$

we may rewrite the above equation as

$$SS_{\mathrm{Res}} = \mathbf{y}'\mathbf{y} - \frac{\left(\sum_{i=1}^{n} y_i\right)^2}{n} - \left[\hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y} - \frac{\left(\sum_{i=1}^{n} y_i\right)^2}{n}\right] \tag{3.12}$$

Therefore, the regression sum of squares is

$$SS_R = \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y} - \frac{\left(\sum_{i=1}^{n} y_i\right)^2}{n} \tag{3.13}$$

the total sum of squares is

$$SS_{\mathrm{T}} = \mathbf{y}'\mathbf{y} - \frac{\left(\sum_{i=1}^{n} y_i\right)^2}{n} \tag{3.14}$$

3.5.2 Tests on Individual Regression Coeficients and Subsets of Coeficients

Once we have determined that at least one of the regressors is important, a logical question becomes which one(s). Adding a variable to a regression model always causes the sum of squares for regression to increase and the residual sum of squares to decrease. We must decide whether the increase in the regression sum of squares is suficient to warrant using the additional regressor in the model. The addition of a regressor also increases the variance of the fitted value , so we must be careful to include only regressors that are of real value in explaining the response. Furthermore, adding an unimportant regressor may increase the residual mean square, which may decrease the usefulness of the model.
The hypotheses for testing the significance of any individual regression coeficient, such as β_j are

$$H_0 : \beta_j = 0; H_1 : \beta_j \neq 0$$

If $H_0 : \beta_j = 0$ is not rejected, then this indicates that the regressor xj can be deleted from the model.

3.6 Prediction of New Observations

The regression model can be used to predict future observations on y corresponding to particular values of the regressor variables, for example, $x_{01}, x_{02}, \ldots, x_{0k}$. If $\boldsymbol{x}_0' = [1, x_{01}, x_{02}, \ldots, x_{0k}]$, then a point estimate of the future observation y_0 at the point $x_{01}, x_{02}, \ldots, x_{0k}$ is

$$\hat{y}_0 = \mathbf{x}_0'\hat{\beta}$$

A $100(1-\alpha)$ percent prediction interval for this future observation is

$$\hat{y}_0 - t_{\alpha/2,n-p}\sqrt{\hat{\sigma}^2\left(1 + \mathbf{x}_0'\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{x}_0\right)} \leq y_0 \leq \hat{y}_0 + t_{\alpha/2,n-p}\sqrt{\hat{\sigma}^2\left(1 + \mathbf{x}_0'\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{x}_0\right)}$$

This is a generalization of the prediction interval for a future observation in simple linear regression